

SECTION 8: Limitations and Challenges

As we explore the capabilities of generative AI, it is equally critical to understand its limitations. This section covers the "dark side" of AI: hallucinations, bias, technical constraints, and ethical dilemmas. Understanding these challenges is essential for responsible and effective use.

Lecture 8.1: Hallucinations and Factual Accuracy

What is an AI Hallucination?

A **hallucination** occurs when a generative model confidently produces content that is factually incorrect, nonsensical, or entirely fabricated.

Key Insight: The AI is not "lying" because it has no concept of truth. It is simply predicting the most probable next word based on patterns, prioritizing fluency over accuracy.

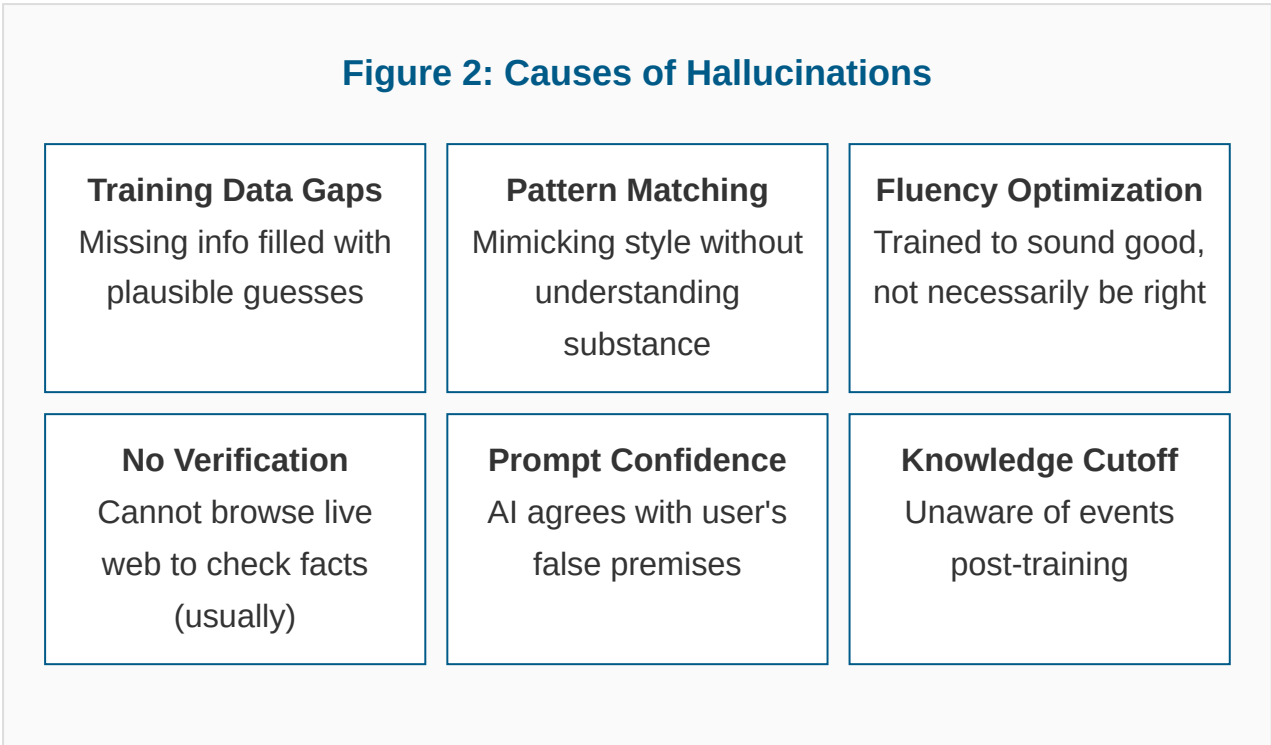
Examples of Hallucinations

- **Factual Errors:** "The Eiffel Tower was built in 1920" (Reality: 1889).
- **Made-up Citations:** Citing non-existent studies by real researchers.
- **Nonsensical Info:** Claims that violate laws of physics.
- **Fabricated Details:** Inventing specific dates or names in a biography.

Figure 1: Hallucination vs. Reality

AI HALLUCINATION "Dr. Sarah Johnson won the 2024 Nobel Prize in Physics for quantum entanglement stability." <i>(Sounds authoritative, detailed, plausible.)</i>	REALITY No such prize was awarded to a Dr. Sarah Johnson. The details are a statistical collage of typical Nobel announcements.
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Why Do Hallucinations Happen?



Types of Hallucinations

Type	Description	Example
Factual	Wrong specific details	Incorrect dates, capitals, or names.
Reasoning	Logical errors	"A is bigger than B, B is bigger than C, so C is bigger than A."
Contextual	Mixing concepts	Conflating two different people with the same name.
Creative	Invented narrative	Adding fictional dialogue to a historical account.

High-Risk Scenarios

Hallucinations are more likely when:

- Asking about **obscure topics** with little training data.
- Requesting **specific details** (quotes, citations, numbers).
- Discussing **recent events** past the knowledge cutoff.
- Using **leading questions** ("Why did X happen?" when X didn't happen).

Detection and Mitigation

Detection Checklist

- ☐ Cross-reference with Google/Wikipedia
- ☐ Be skeptical of unsourced specificity
- ☐ Verify all citations/links
- ☐ Check consistency (ask multiple times)

Mitigation Prompts

- "If you don't know, explicitly say 'I don't know'."
- "Quote directly from the provided text only."
- "Rate your confidence in this answer."
- Use **RAG** (Retrieval-Augmented Generation) to ground answers.

The Confidence-Accuracy Gap

AI models often express high confidence even when they are completely wrong. Never use "tone of voice" as a proxy for truthfulness.

LECTURE 8.1 KEY TAKEAWAY

Hallucinations are a feature, not a bug, of how LLMs work. They prioritize plausible text generation over factual accuracy. Users must always verify high-stakes information.

Lecture 8.2: Bias and Fairness Issues

What is Bias in AI?

Bias in AI refers to systematic, unfair discrimination in outputs based on characteristics like race, gender, age, or religion. It is not just a technical glitch; it causes real-world social harm.

Sources of Bias

Figure 3: Where Bias Comes From

Training Data (The Mirror)

Historical prejudice, representation gaps, and stereotypes present in the internet text used for training.

Algorithms (The Lens)

Optimization choices, feature selection, and feedback loops that amplify existing data biases.

Manifestations of Bias

- **Stereotyping:** Associating "Nurse" with women and "Engineer" with men.
- **Representation:** Generating images of "CEOs" that are almost exclusively white men.
- **Quality Disparity:** Speech recognition working poorly for certain accents or dialects.
- **Toxicity:** Models being more likely to generate toxic comments about marginalized groups.

Real-World Impacts

Economic Harm

Biased hiring tools filtering out qualified candidates; biased credit scoring denying loans.

Social Harm

Reinforcing harmful stereotypes and marginalizing groups in media and culture.

Epistemic Harm

Distributing misinformation that disproportionately affects certain communities.

Dignitary Harm

The psychological impact of being invisible, misrepresented, or demeaned by technology.

Addressing Bias

Level	Interventions
Data	Diverse collection, balancing datasets, filtering toxic content.
Algorithm	Fairness constraints, adversarial debiasing, RLHF tuning.
Evaluation	Red-teaming, bias benchmarking, user feedback loops.

Bias is Inevitable but Manageable

Complete neutrality is impossible because data reflects human history. However, we can measure, mitigate, and manage bias through continuous improvement and human oversight.

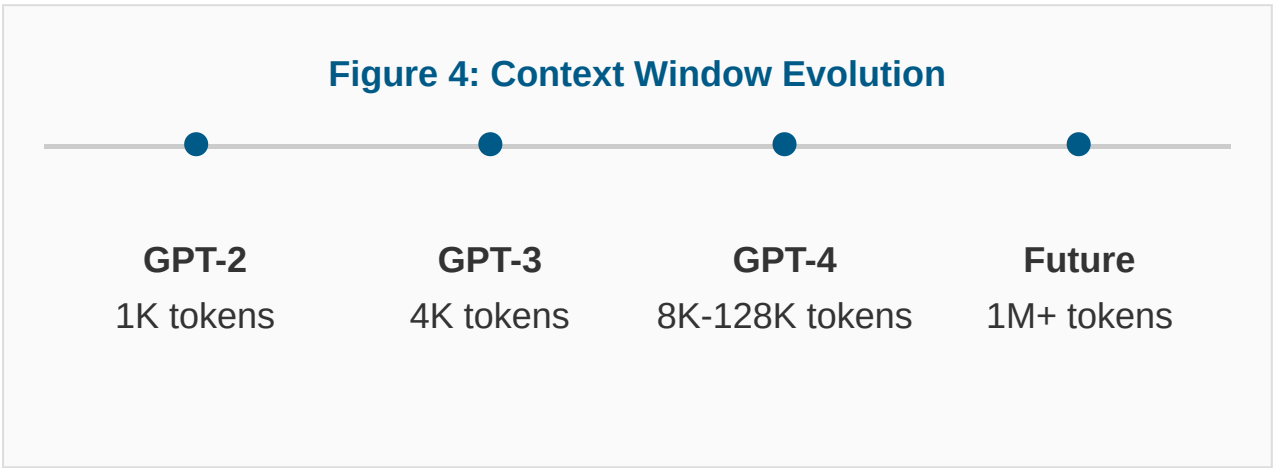
LECTURE 8.2 KEY TAKEAWAY

Bias originates from the data we feed the models. While technical fixes help, user awareness and critical evaluation are the ultimate safeguards against unfair discrimination.

Lecture 8.3: Technical Limitations

1. Context Window Constraints

The **Context Window** is the limit on how much text a model can process at once (input + output). If a conversation exceeds this limit, the model "forgets" earlier parts.



2. Memory and Reasoning

- **No Long-Term Memory:** Models do not remember you between sessions unless designed with a separate database.

- **Reasoning Depth:** They struggle with long, multi-step logical chains and often lose the thread.
- **Working Memory:** Tracking many variables simultaneously (e.g., a complex logic puzzle) is error-prone.

3. Mathematical & Logical Limitations

LLMs are language engines, not calculators. They predict the next token in a math problem rather than computing it.

Common Math Errors

Input: "What is $4,729 * 863$?"
AI Output: "4,081,127" (Incorrect)
Actual: 4,081,127 is a hallucinated number.

4. Reliability Issues

- **Stochasticity:** The same prompt can yield different results each time.
- **Drift:** Quality often degrades in very long generations.
- **Sensitivity:** Small changes in wording can radically change the output.

5. Knowledge Cutoff

Models have static knowledge. They do not know about events that happened after their training data cutoff unless connected to live search tools.

Understanding Limitations is Empowering

Knowing what AI *cannot* do prevents frustration and failure. Use AI for drafting, ideation, and summarizing, but not for unverified math, high-stakes logic, or current events without search.

LECTURE 8.3 KEY TAKEAWAY

Generative AI is not magic. It is constrained by compute, context memory, and its fundamental nature as a probabilistic token predictor, not a reasoning engine.

Lecture 8.4: Ethical and Safety Concerns

1. Misinformation & Disinformation

Misinformation is unintentional falsehood; **Disinformation** is deliberate deception. AI scales both.

- **Deepfakes:** Realistic video/audio impersonations.
- **Synthetic Media:** Fake news articles generated at scale.
- **Erosion of Trust:** When anything can be faked, believing reality becomes harder.

2. Copyright & Intellectual Property

Models are trained on billions of copyrighted works (books, art, code).

Legal Questions Unsettled

Is training "Fair Use"? Who owns the output? Can an AI infringe on style? These legal battles are ongoing as of 2025.

3. Privacy Concerns

- **Data Leakage:** Training data might contain private info (PII) that models memorize and regurgitate.
- **User Privacy:** Prompts sent to cloud models may be stored and reviewed.

4. Labor & Economic Impact

Figure 5: The Economic Impact Spectrum



Displacement & Obsolescence Transition & Adaptation Augmentation & Productivity

AI creates "Positive Possibilities" (productivity, new jobs) but also risks "Displacement" (automation of writing/coding tasks) and "Devaluation" of human skills.

5. Dual-Use Dilemma

The same technology that helps write code can help write malware. The same tool that discovers drugs can discover toxins. Balancing open access with safety is a major challenge.

6. Responsible AI Use

Your Responsibilities as a User

- **Transparency:** Disclose when content is AI-generated.
- **Verification:** Always fact-check before sharing.
- **Respect:** Do not use AI to impersonate or harm others.
- **Feedback:** Report biases and errors to developers.

LECTURE 8.4 KEY TAKEAWAY

With great power comes great responsibility. The ethical use of AI requires vigilance against misinformation, respect for privacy and IP, and a conscious effort to augment—not replace—human agency.